Design, Implementation and Big Data Integration of a Dimensional Data Warehouse for Inventory Management at ABC Consumer Electronics

Student Name:

Student ID:

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1. Introduction

This coursework is aimed at designing, implementing, and evaluating a whole data warehousing and Big Data solution for ABC Consumer Electronics Outlet Ltd., a multi-channel retail organization that has outlets in the UK and Europe. The firm is now employing a range of independent software, including Vend, Linnworks, and Xerox, to take control of the retail, online transactions, and financial processes. Despite the fact that these systems facilitate day-to-day operations, they create extensive amounts of data that are not linked together, and as such, it becomes challenging to present integrated reports and data-led insights by the managers. This coursework will deal with these challenges using a single data warehouse solution.

The project is specifically in the inventory management business unit of the organization, which involves mailing purchase orders, receiving stock, and stocking the stock of the product. These steps allow preventing overstocking and understocking, which directly affect the revenue, operational efficiency, and customer satisfaction (Nambiar and Mundra, 2022). The data warehouse will facilitate analysis, including current stock levels, low stock items per week, and how each supplier performed in orders by building well-constructed dimensional models and using star schema to enable it.

The project mostly aims at the inventory process, which constitutes ordering of purchase orders, stock receipt, and daily stock balances. These processes play a crucial role in avoiding a problem of overstocking and stock-outs, which have direct impacts on customer satisfaction and efficiency of the business. The second step undertaken after implementing and testing the warehouse in the Oracle Live SQL was to migrate the exported warehouse data into a big data environment with the help of Google Colab. In this case, the Apache Pig was used to load, manipulate, and analyze the CSV export (Karamolegkos et al., 2023). This represents the collaboration of a standard SQL-based warehouse and the current Big Data processing solutions to optimize strategic decision-making.

2. Analysis & Design of Dimensional Data Model

2.1 Identification and Definition of the Grain for the Three Central Fact Tables

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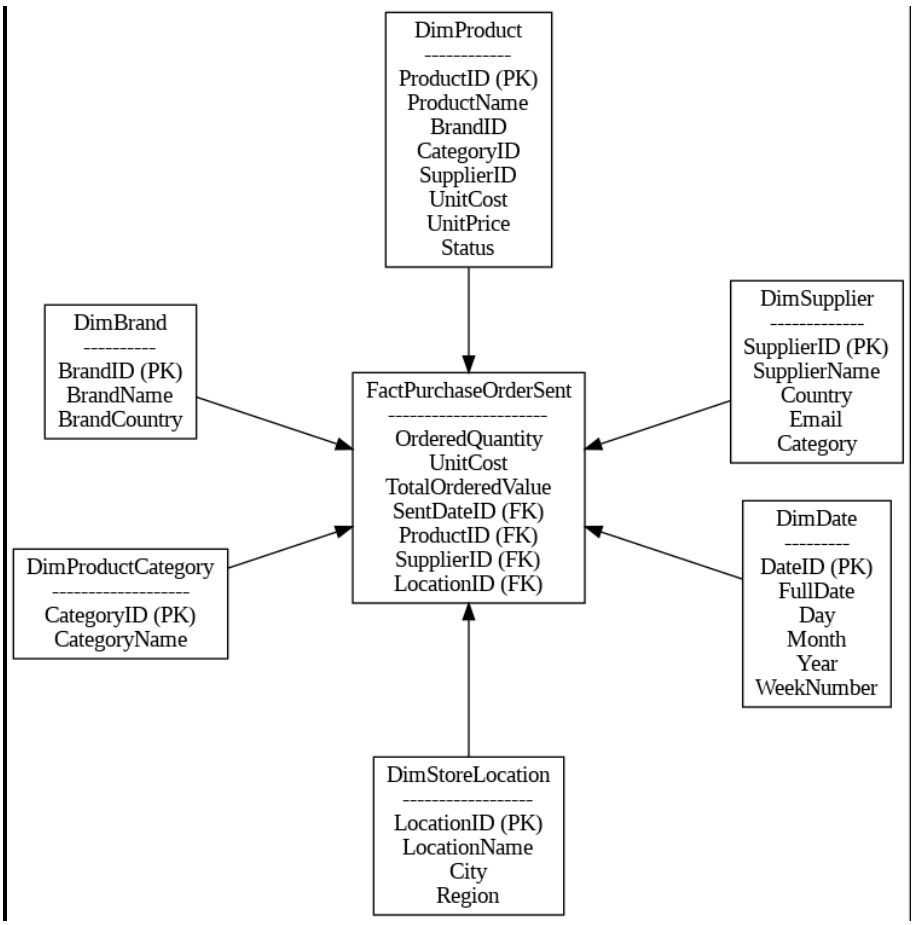

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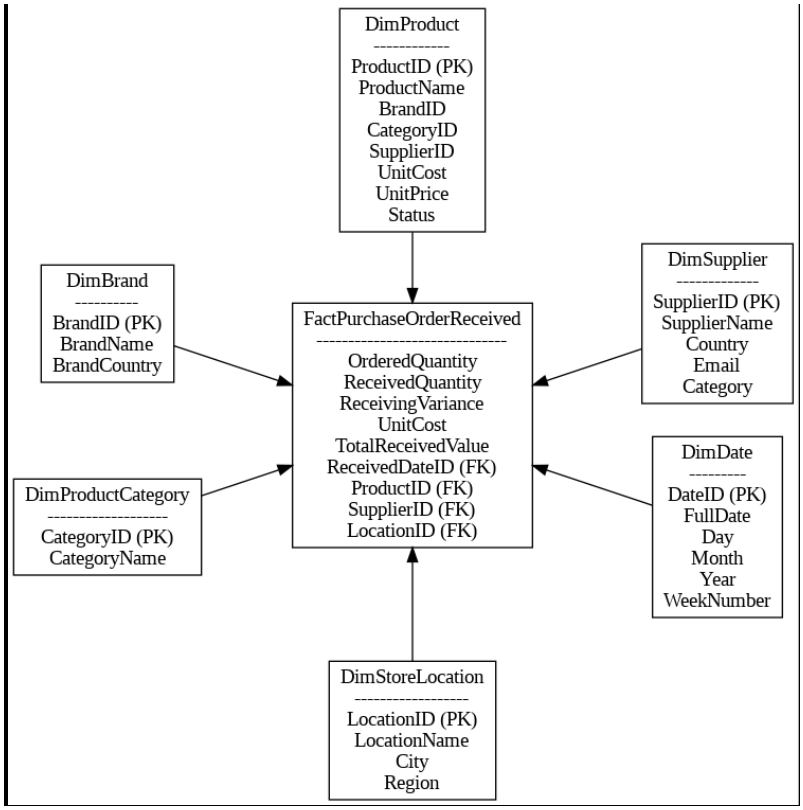

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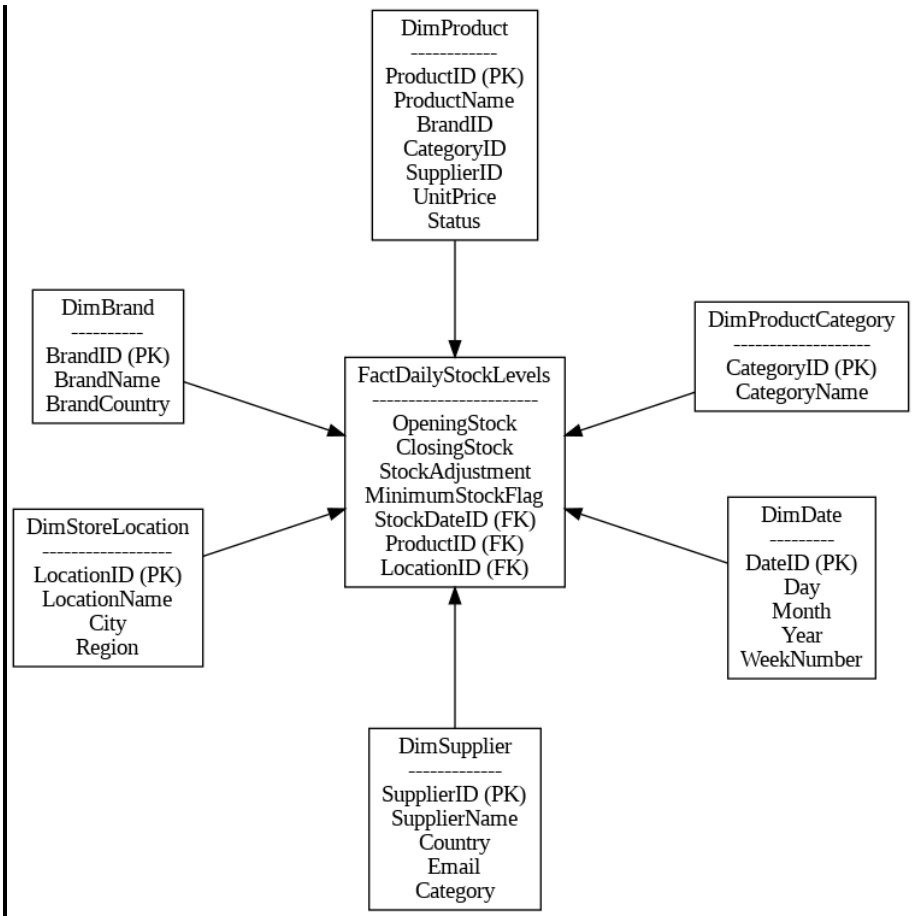

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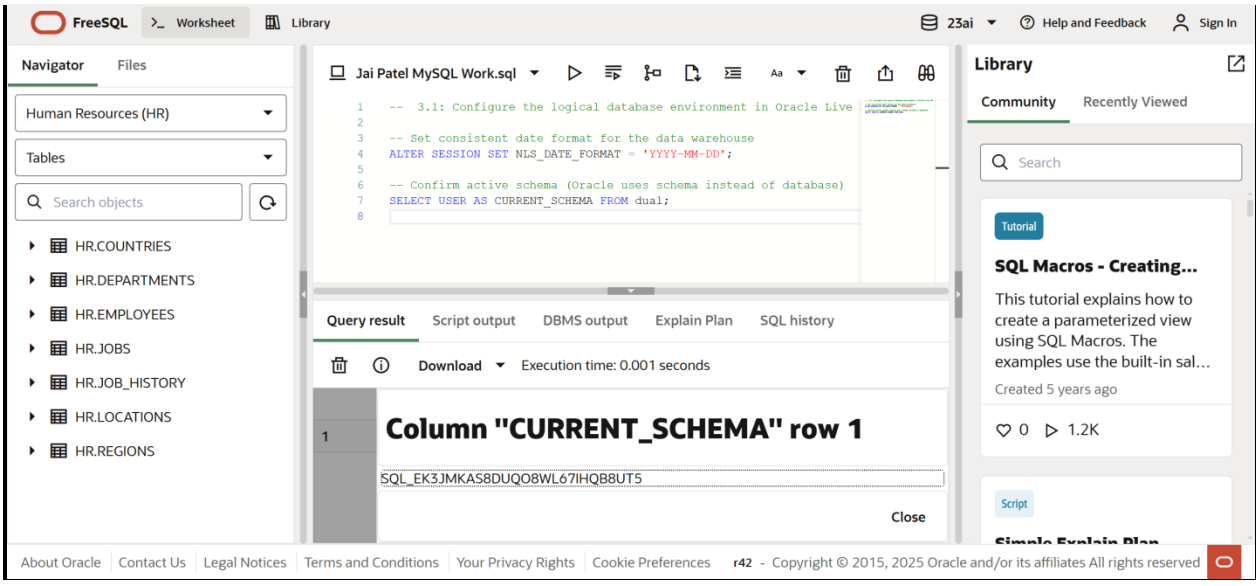
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HR.JOB_HISTORY
HR.LOCATIONS
HR.REGIONS
Jai Patel MySQL Work.sql
-- 3.1: Configure the logical database environment in Oracle Live
-- Set consistent date format for the data warehouse
ALTER SESSION SET NLS_DATE_FORMAT = 'YYYY-MM-DD';
-- Confirm active schema (Oracle uses schema instead of database)
SELECT USER AS CURRENT_SCHEMA FROM dual;
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.001 seconds
Column "CURRENT_SCHEMA" row 1
SQL_EK3JMKAS8DUQO8WL67IHQB8UT5
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This tutorial explains how to create a parameterized view using SQL Macros. The examples use the built-in sal...
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Jai Patel MySQL Work.sql

```sql
-- 3.2: Create Dimension Tables

-- Date Dimension
CREATE TABLE DimDate (
    DateID       NUMBER PRIMARY KEY,
    FullDate     DATE,
    DayNumber    NUMBER,
    MonthNumber  NUMBER,
    QuarterNumber NUMBER,
    YearNumber   NUMBER,
    WeekNumber   NUMBER,
    DayOfWeek    VARCHAR2(20)
);

-- Supplier Dimension
CREATE TABLE DimSupplier (
    SupplierID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    SupplierName      VARCHAR2(200),
    SupplierAddress   VARCHAR2(300),
    SupplierPhone     VARCHAR2(50),
    SupplierEmail     VARCHAR2(150),
    Country           VARCHAR2(100),
    SupplierCategory  VARCHAR2(100)
);

-- Brand Dimension
CREATE TABLE DimBrand (
    BrandID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    BrandName      VARCHAR2(200),
    BrandCountry   VARCHAR2(100),
    ParentCompany  VARCHAR2(200)
);
```

Jai Patel MySQL Work.sql

```sql
CREATE TABLE DimProductCategory (
    CategoryID          NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    CategoryName        VARCHAR2(200),
    CategoryDescription VARCHAR2(300),
    Department          VARCHAR2(200)
);

-- Store / Location Dimension
CREATE TABLE DimStoreLocation (
    LocationID    NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    LocationName  VARCHAR2(200),
    Address       VARCHAR2(300),
    City          VARCHAR2(100),
    Region        VARCHAR2(100),
    Country       VARCHAR2(100),
    StoreType     VARCHAR2(50)
);

-- Product Dimension
CREATE TABLE DimProduct (
    ProductID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    ProductName      VARCHAR2(200),
    SKU              VARCHAR2(100),
    BrandID          NUMBER,
    CategoryID       NUMBER,
    SupplierID       NUMBER,
    UnitCost         NUMBER(10,2),
    UnitPrice        NUMBER(10,2),
    ProductDescription VARCHAR2(300),
    Status           VARCHAR2(50)
);
```

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My Schema

Tables

Search objects

- AGENT
- APPOINTMENT
- AUTHORS
- BOOKS
- CATEGORIES
- CUSTOMER
- DIMBRAND
- DIMDATE
- DIMPRODUCT
- DIMPRODUCTCATEGORY
- DIMSTORELOCATION
- DIMSUPPLIER
- LIBRARIANS
- LOANS
- MEMBERS

Jai Patel MySQL Work.sql

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE TABLE DimDate (
        DateID       NUMBER PRIMARY KEY,
        FullDate     DATE,
        DayNumber    NUMBER,...
Show more...


Table DIMDATE created.

Elapsed: 00:00:00.019

SQL> CREATE TABLE DimSupplier (
        SupplierID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
        SupplierName      VARCHAR2(200),
        SupplierAddress   VARCHAR2(300),...
Show more...


Table DIMSUPPLIER created.

Elapsed: 00:00:00.016

SQL> CREATE TABLE DimBrand (
        BrandID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
        BrandName      VARCHAR2(200),
        BrandCountry   VARCHAR2(100),...
Show more...
```

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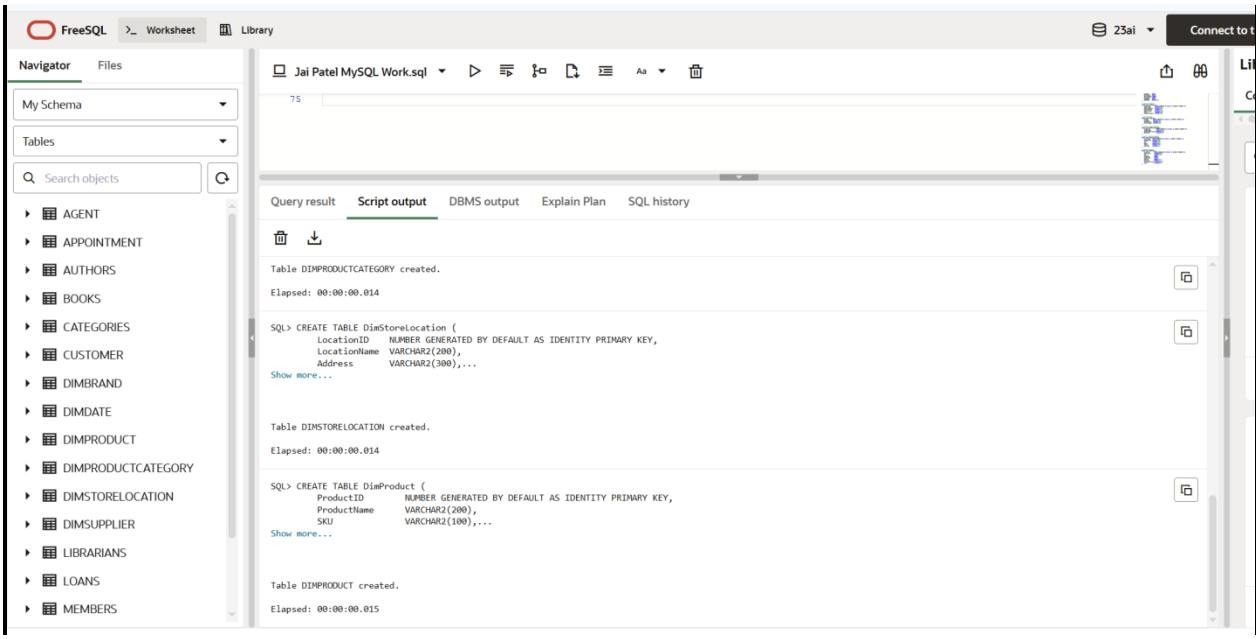

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Jai Patel MySQL Work.sql*

```sql

-- Fact: Purchase Orders Sent
CREATE TABLE FactPurchaseOrderSent (
    SentID                NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    PurchaseOrderNumber   VARCHAR2(100),
    SentDateID            NUMBER,
    ProductID             NUMBER,
    SupplierID            NUMBER,
    LocationID            NUMBER,
    OrderedQuantity       NUMBER,
    UnitCost              NUMBER(10,2),
    TotalOrderedValue     NUMBER(12,2)
);

-- Fact: Purchase Orders Received
CREATE TABLE FactPurchaseOrderReceived (
    ReceivedID            NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    PurchaseOrderNumber   VARCHAR2(100),
    ReceivedDateID        NUMBER,
    ProductID             NUMBER,
    SupplierID            NUMBER,
    LocationID            NUMBER,
    OrderedQuantity       NUMBER,
    ReceivedQuantity      NUMBER,
    ReceivingVariance     NUMBER,
    UnitCost              NUMBER(10,2),
    TotalReceivedValue    NUMBER(12,2)
);

-- Fact: Daily Stock Levels
CREATE TABLE FactDailyStockLevels (
    StockLevelID        NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    StockDateID         NUMBER,
    ProductID           NUMBER,
    LocationID          NUMBER,
    OpeningStock        NUMBER,
    ClosingStock        NUMBER,
    StockAdjustment     NUMBER,
    MinimumStockFlag    VARCHAR2(10)
);
```

[redacted]

Jai Patel MySQL Work.sql

```sql

-- Fact: Purchase Orders Sent
CREATE TABLE FactPurchaseOrderSent (
    SentID                NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    PurchaseOrderNumber   VARCHAR2(100),
    SentDateID            NUMBER,
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Show more...

Table FACTPURCHASEORDERSENT created.

Elapsed: 00:00:00.030

SQL> CREATE TABLE FactPurchaseOrderReceived (
        ReceivedID           NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
        PurchaseOrderNumber  VARCHAR2(100),
        ReceivedDateID       NUMBER,...
Show more...

Table FACTPURCHASEORDERRECEIVED created.

Elapsed: 00:00:00.016

SQL> CREATE TABLE FactDailyStockLevels (
        StockLevelID       NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
        StockDateID        NUMBER,
        ProductID          NUMBER,...
Show more...
```

[redacted]

Jai Patel MySQL Work.sql

```sql

-- 3.4: Add Foreign Key Constraints

-- FactPurchaseOrderSent Dimensions
ALTER TABLE FactPurchaseOrderSent
ADD CONSTRAINT fk_fpos_date
FOREIGN KEY (SentDateID) REFERENCES DimDate(DateID);

ALTER TABLE FactPurchaseOrderSent
ADD CONSTRAINT fk_fpos_product
FOREIGN KEY (ProductID) REFERENCES DimProduct(ProductID);

ALTER TABLE FactPurchaseOrderSent
ADD CONSTRAINT fk_fpos_supplier
FOREIGN KEY (SupplierID) REFERENCES DimSupplier(SupplierID);

ALTER TABLE FactPurchaseOrderSent
ADD CONSTRAINT fk_fpos_location
FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID);



-- FactPurchaseOrderReceived Dimensions
ALTER TABLE FactPurchaseOrderReceived
ADD CONSTRAINT fk_fpor_date
FOREIGN KEY (ReceivedDateID) REFERENCES DimDate(DateID);

ALTER TABLE FactPurchaseOrderReceived
ADD CONSTRAINT fk_fpor_product
FOREIGN KEY (ProductID) REFERENCES DimProduct(ProductID);

ALTER TABLE FactPurchaseOrderReceived
ADD CONSTRAINT fk_fpor_supplier
FOREIGN KEY (SupplierID) REFERENCES DimSupplier(SupplierID);

ALTER TABLE FactPurchaseOrderReceived
ADD CONSTRAINT fk_fpor_location
FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID);

```

```sql
-- FactDailyStockLevels Dimensions
ALTER TABLE FactDailyStockLevels
ADD CONSTRAINT fk_fdsl_date
FOREIGN KEY (StockDateID) REFERENCES DimDate(DateID);

ALTER TABLE FactDailyStockLevels
ADD CONSTRAINT fk_fdsl_product
FOREIGN KEY (ProductID) REFERENCES DimProduct(ProductID);

ALTER TABLE FactDailyStockLevels
ADD CONSTRAINT fk_fdsl_location
FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID);
```

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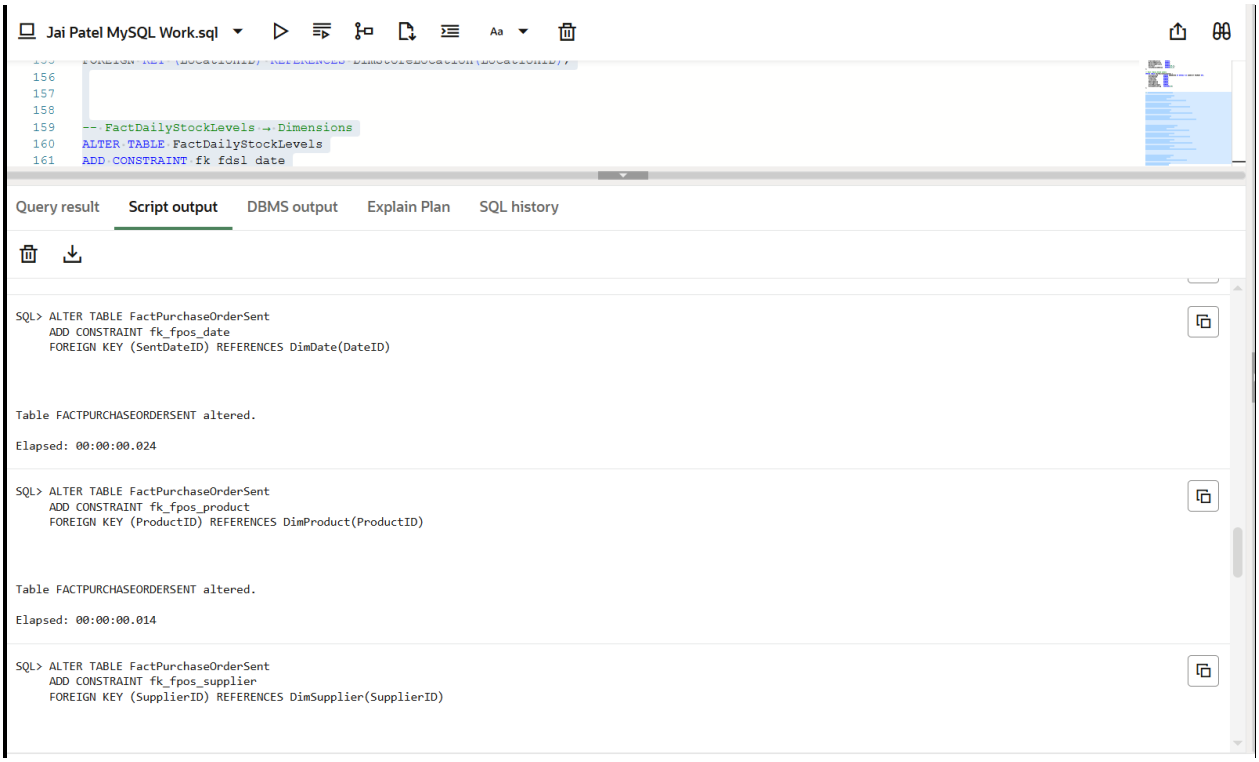

[redacted]

Jai Patel MySQL Work.sql

```sql
-- FactDailyStockLevels → Dimensions
ALTER TABLE FactDailyStockLevels
ADD CONSTRAINT fk_fdsl_date
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Table FACTPURCHASEORDERSENT altered.

Elapsed: 00:00:00.014

SQL> ALTER TABLE FactPurchaseOrderSent
     ADD CONSTRAINT fk_fpos_location
     FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID)


Table FACTPURCHASEORDERSENT altered.

Elapsed: 00:00:00.013

SQL> ALTER TABLE FactPurchaseOrderReceived
     ADD CONSTRAINT fk_fpor_date
     FOREIGN KEY (ReceivedDateID) REFERENCES DimDate(DateID)


Table FACTPURCHASEORDERRECEIVED altered.

Elapsed: 00:00:00.016

SQL> ALTER TABLE FactPurchaseOrderReceived
     ADD CONSTRAINT fk_fpor_product
     FOREIGN KEY (ProductID) REFERENCES DimProduct(ProductID)
```

[redacted]

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Table FACTPURCHASEORDERRECEIVED altered.

Elapsed: 00:00:00.013

SQL> ALTER TABLE FactPurchaseOrderReceived
     ADD CONSTRAINT fk_fpor_supplier
     FOREIGN KEY (SupplierID) REFERENCES DimSupplier(SupplierID)


Table FACTPURCHASEORDERRECEIVED altered.

Elapsed: 00:00:00.012

SQL> ALTER TABLE FactPurchaseOrderReceived
     ADD CONSTRAINT fk_fpor_location
     FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID)


Table FACTPURCHASEORDERRECEIVED altered.

Elapsed: 00:00:00.011
```

[redacted]

Query result | Script output | DBMS output | Explain Plan | SQL history

```
     FOREIGN KEY (StockDateID) REFERENCES DimDate(DateID)


Table FACTDAILYSTOCKLEVELS altered.

Elapsed: 00:00:00.014

SQL> ALTER TABLE FactDailyStockLevels
     ADD CONSTRAINT fk_fdsl_product
     FOREIGN KEY (ProductID) REFERENCES DimProduct(ProductID)


Table FACTDAILYSTOCKLEVELS altered.

Elapsed: 00:00:00.013

SQL> ALTER TABLE FactDailyStockLevels
     ADD CONSTRAINT fk_fdsl_location
     FOREIGN KEY (LocationID) REFERENCES DimStoreLocation(LocationID)


Table FACTDAILYSTOCKLEVELS altered.

Elapsed: 00:00:00.013
```

[redacted]

Jai Patel MySQL Work.sql*

```sql
(20241209,9,9,10,18,8,'Yes'),
(20241210,10,10,5,12,7,'Yes');

--- 3.5.1 implementation and testing of the data warehouse
SELECT 'DimDate' AS TableName, COUNT(*) AS TotalRows FROM DimDate
UNION ALL SELECT 'DimSupplier', COUNT(*) FROM DimSupplier
UNION ALL SELECT 'DimBrand', COUNT(*) FROM DimBrand
UNION ALL SELECT 'DimProductCategory', COUNT(*) FROM DimProductCategory
UNION ALL SELECT 'DimStoreLocation', COUNT(*) FROM DimStoreLocation
UNION ALL SELECT 'DimProduct', COUNT(*) FROM DimProduct
UNION ALL SELECT 'FactPurchaseOrderSent', COUNT(*) FROM FactPurchaseOrderSent
UNION ALL SELECT 'FactPurchaseOrderReceived', COUNT(*) FROM FactPurchaseOrderReceived
UNION ALL SELECT 'FactDailyStockLevels', COUNT(*) FROM FactDailyStockLevels;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.153 seconds

	TABLENAME	TOTALROWS
1	DimDate	10
2	DimSupplier	10
3	DimBrand	10
4	DimProductCategory	10
5	DimStoreLocation	10
6	DimProduct	10
7	FactPurchaseOrderSe	10
8	FactPurchaseOrderRe	10
9	FactDailyStockLevels	10

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[redacted]

Jai Patel MySQL Work.sql

```sql
-- 3.5.2 Check Referential Integrity (JOIN all dimensions with a fact)
SELECT
    f.PurchaseOrderNumber,
    d.FullDate,
    p.ProductName,
    s.SupplierName,
    l.LocationName,
    f.OrderedQuantity,
    f.TotalOrderedValue
FROM FactPurchaseOrderSent f
JOIN DimDate d ON f.SentDateID = d.DateID
JOIN DimProduct p ON f.ProductID = p.ProductID
JOIN DimSupplier s ON f.SupplierID = s.SupplierID
JOIN DimStoreLocation l ON f.LocationID = l.LocationID
FETCH FIRST 10 ROWS ONLY;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.185 seconds

	PURCHASEORDERN	FULLDATE	PRODUCTNAME	SUPPLIERNAME	LOCATIONNAME	ORDEREDQUANTIT	TOTALORDEREDVAI
1	PO1001	12/1/2024, 12:00:0(	Samsung 55 Inch LEI	TechSource Ltd	Central Warehouse	20	6000
2	PO1002	12/2/2024, 12:00:0(	Sony Wireless Headp	Global Electronics	Oxford Street Store	50	2500
3	PO1003	12/3/2024, 12:00:0(	LG 4K OLED TV	SmartSupply Co	Manchester Store	10	6500
4	PO1004	12/4/2024, 12:00:0(	Dell Inspiron Laptop	EuroTech Traders	Birmingham Store	15	6000
5	PO1005	12/5/2024, 12:00:0(	HP Laser Printer	MegaWarehouse	Liverpool Store	25	3000
6	PO1006	12/6/2024, 12:00:0(	Lenovo ThinkPad	FutureElectro	Leeds Warehouse	18	9000
7	PO1007	12/7/2024, 12:00:0(	JBL Bluetooth Speak	BrightLine	Bristol Store	40	1000
8	PO1008	12/8/2024, 12:00:0(	Bose Home Speaker	SwiftSuppliers	Nottingham Store	12	1800
9	PO1009	12/9/2024, 12:00:0(	Canon DSLR Camera	TechOne UK	Newcastle Store	8	2800

[redacted]

Jai Patel MySQL Work.sql

```sql
-- 3.5.2 Daily Minimum Stock Report (Business Requirement)
SELECT
    d.FullDate,
    p.ProductName,
    l.LocationName,
    f.ClosingStock,
    f.MinimumStockFlag
FROM FactDailyStockLevels f
JOIN DimDate d ON f.StockDateID = d.DateID
JOIN DimProduct p ON f.ProductID = p.ProductID
JOIN DimStoreLocation l ON f.LocationID = l.LocationID
WHERE f.MinimumStockFlag = 'Yes';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.019 seconds

	FULLDATE	PRODUCTNAME	LOCATIONNAME	CLOSINGSTOCK	MINIMUMSTOCKFL
1	12/9/2024, 12:00:0(	Canon DSLR Camera	Newcastle Store	18	Yes
2	12/10/2024, 12:00:(	Apple iPhone 14	Sheffield Store	12	Yes

[redacted]

Jai Patel MySQL Work.sql

```sql
--3.5.4 Supplier Delivery Performance (Ordered vs Received)

SELECT
    r.PurchaseOrderNumber,
    p.ProductName,
    s.SupplierName,
    r.OrderedQuantity,
    r.ReceivedQuantity,
    r.ReceivingVariance
FROM FactPurchaseOrderReceived r
JOIN DimProduct p ON r.ProductID = p.ProductID
JOIN DimSupplier s ON r.SupplierID = s.SupplierID;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.019 seconds

	PURCHASEORDERN	PRODUCTNAME	SUPPLIERNAME	ORDEREDQUANTIT	RECEIVEDQUANTIT	RECEIVINGVARIANC
1	PO1001	Samsung 55 Inch LEI	TechSource Ltd	20	20	0
2	PO1002	Sony Wireless Headp	Global Electronics	50	48	-2
3	PO1003	LG 4K OLED TV	SmartSupply Co	10	10	0
4	PO1004	Dell Inspiron Laptop	EuroTech Traders	15	15	0
5	PO1005	HP Laser Printer	MegaWarehouse	25	24	-1
6	PO1006	Lenovo ThinkPad	FutureElectro	18	18	0
7	PO1007	JBL Bluetooth Speak	BrightLine	40	39	-1
8	PO1008	Bose Home Speaker	SwiftSuppliers	12	12	0
9	PO1009	Canon DSLR Camera	TechOne UK	8	8	0

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[redacted]

```sql
--3.5.5 Total Stock Movement by Product
SELECT
    p.ProductName,
    SUM(f.OpeningStock) AS TotalOpening,
    SUM(f.ClosingStock) AS TotalClosing,
    SUM(f.StockAdjustment) AS Adjustments
FROM FactDailyStockLevels f
JOIN DimProduct p ON f.ProductID = p.ProductID
GROUP BY p.ProductName;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.016 seconds

	PRODUCTNAME	TOTALOPENING	TOTALCLOSING	ADJUSTMENTS
1	Samsung 55 Inch LEI	50	70	20
2	Sony Wireless Headp	100	150	50
3	LG 4K OLED TV	30	40	10
4	Dell Inspiron Laptop	20	35	15
5	HP Laser Printer	60	85	25
6	Lenovo ThinkPad	25	43	18
7	JBL Bluetooth Speak	200	240	40
8	Bose Home Speaker	15	27	12
9	Canon DSLR Camera	10	18	8

[redacted]

Jai Patel MySQL Work.sql

```sql
GROUP BY p.ProductName;

--3.5.6 Brand-Level Purchase Analysis
SELECT
    b.BrandName,
    SUM(f.TotalOrderedValue) AS TotalValueOrdered
FROM FactPurchaseOrderSent f
JOIN DimProduct p ON f.ProductID = p.ProductID
JOIN DimBrand b ON p.BrandID = b.BrandID
GROUP BY b.BrandName;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download | Execution time: 0.024 seconds

	BRANDNAME	TOTALVALUEORDER
1	Samsung	6000
2	Sony	23500
3	LG	6500
4	Panasonic	2800
5	Dell	6000
6	HP	3000
7	Lenovo	9000
8	Bose	1800
9	JBL	1000

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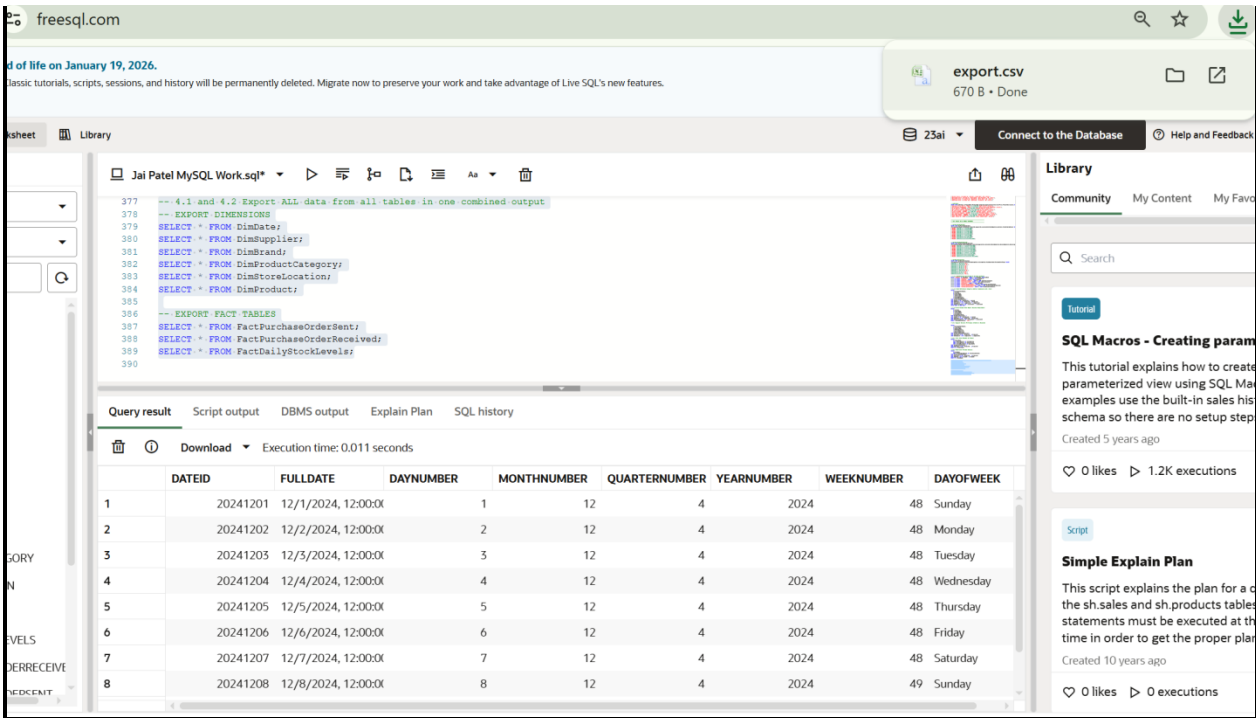

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```
[1]
1m
!apt-get install openjdk-8-jdk-headless -qq
!wget https://archive.apache.org/dist/hadoop/core/hadoop-3.2.2/hadoop-3.2.2.tar.gz
!tar -xzf hadoop-3.2.2.tar.gz
!mv hadoop-3.2.2 /usr/local/hadoop

!wget https://archive.apache.org/dist/pig/pig-0.17.0/pig-0.17.0.tar.gz
!tar -xzf pig-0.17.0.tar.gz
!mv pig-0.17.0 /usr/local/pig
```

```
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/jsadebugd to provide /usr/bin/jsadebugd (jsadebugd) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/native2ascii to provide /usr/bin/native2ascii (native2ascii) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/rmic to provide /usr/bin/rmic (rmic) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/schemagen to provide /usr/bin/schemagen (schemagen) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/wsgen to provide /usr/bin/wsgen (wsgen) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/wsimport to provide /usr/bin/wsimport (wsimport) in auto mode
update-alternatives: using /usr/lib/jvm/java-8-openjdk-amd64/bin/xjc to provide /usr/bin/xjc (xjc) in auto mode
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link
```

[redacted]

```bash
%%bash
cat > /usr/local/hadoop/etc/hadoop/core-site.xml <<EOF
<configuration>
 <property>
   <name>fs.defaultFS</name>
   <value>hdfs://localhost:9000</value>
 </property>
</configuration>
EOF
```

```bash
%%bash
cat > /usr/local/hadoop/etc/hadoop/hdfs-site.xml <<EOF
<configuration>
 <property>
   <name>dfs.replication</name>
   <value>1</value>
 </property>
 <property>
   <name>dfs.namenode.name.dir</name>
   <value>/usr/local/hadoop/hdfs/namenode</value>
 </property>
 <property>
   <name>dfs.datanode.data.dir</name>
   <value>/usr/local/hadoop/hdfs/datanode</value>
 </property>
</configuration>
EOF
```

```bash
!mkdir -p /content/hdfs/data
!cp export.csv /content/hdfs/data/
```

[redacted]

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```bash
%%bash
cat << 'EOF' > stock_analysis.pig

-- Load CSV from the pseudo-HDFS folder
raw_data = LOAD '/content/hdfs/data/export.csv'
USING PigStorage(',')
AS (
    DateID:int,
    ProductID:int,
    LocationID:int,
    OpeningStock:int,
    ClosingStock:int,
    StockAdjustment:int,
    MinimumStockFlag:chararray
);

-- Filter minimum stock
min_stock = FILTER raw_data BY MinimumStockFlag == 'Yes';

-- Group by product
grouped = GROUP raw_data BY ProductID;

-- Calculate total adjustments
summary = FOREACH grouped GENERATE
    group AS ProductID,
    SUM(raw_data.StockAdjustment) AS TotalAdjustment;

-- Store output
STORE summary INTO '/content/hdfs/output'
USING PigStorage(',');
EOF
```

[redacted]

```
!/usr/local/pig/bin/pig -x local stock_analysis.pig
```

```
                Reduce output records=1
                Spilled Records=1
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=304087040
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=0
2025-12-12 05:10:20,665 [pool-4-thread-1] INFO  org.apache.hadoop.mapred.LocalJobRunner - Finishing task: attempt_local754143284_0001_r_000000_0
2025-12-12 05:10:20,666 [Thread-6] INFO  org.apache.hadoop.mapred.LocalJobRunner - reduce task executor complete.
2025-12-12 05:10:20,793 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,819 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,822 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2025-12-12 05:10:20,824 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,889 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2025-12-12 05:10:20,895 [main] INFO  org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.2.2   0.17.0  root    2025-12-12 05:10:18     2025-12-12 05:10:20     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_local754143284_0001 1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     grouped,raw_data,summary        GROUP_BY,COMBINER       /content/hdfs/output,

Input(s):
Successfully read 11 records from: "/content/hdfs/data/export.csv"
```

```
Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_local754143284_0001 1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     grouped,raw_data,summary        GROUP_BY,COMBINER       /content/hdfs/output,

Input(s):
Successfully read 11 records from: "/content/hdfs/data/export.csv"

Output(s):
Successfully stored 1 records in: "/content/hdfs/output"

Counters:
Total records written : 1
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local754143284_0001


2025-12-12 05:10:20,899 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,903 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,905 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2025-12-12 05:10:20,914 [main] WARN  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 16 time(s).
2025-12-12 05:10:20,914 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2025-12-12 05:10:20,948 [main] INFO  org.apache.pig.Main - Pig script completed in 5 seconds and 297 milliseconds (5297 ms)
```

```
!cat /content/hdfs/output/part-r-00000
```

```
,20240
```

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